

RANCANGAN APLIKASI CBT PROFESIONAL DAN MENGENAL SUARA HEWAN

Nama: Mohammad Yusuf Fathoni

Nim: 14523051

Kelas: Ti-2 smtr 6

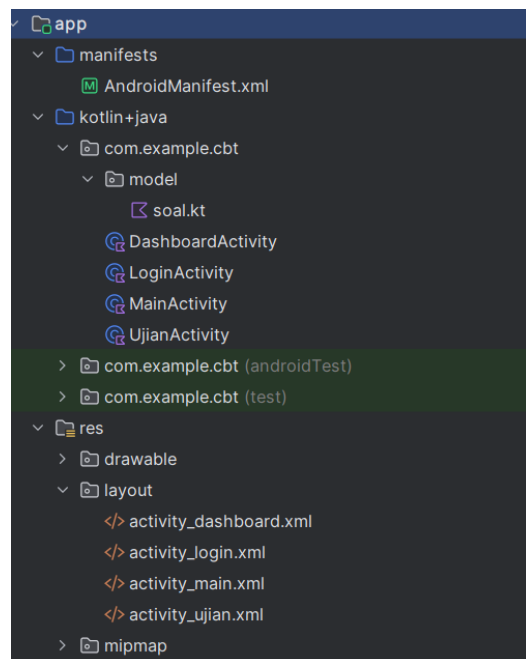
Laporan Praktikum ke-8

RANCANGAN APLIKASI CBT PROFESIONAL

1. Buat Project baru dengan create new project kemudian pilih empty views activity kemudian isi

- Name: cbt
- Packagename: com.example.cbt
- Language: Kotlin
- Klik selesai

kemudian buat struktur folder projectnya seperti ini:



2. kemudian tambahkan program-program di bawah ini:

- **Manifests/AndroidManifest.xml**
-

```

<manifest xmlns:android="http://schemas.android.com/apk/res/android">
    <application
        android:theme="@style/Theme.Cbt" android:allowBackup="false"
        android:usesCleartextTraffic="true" android:label="CBT
        STIKOM">
        <activity
            android:name=".UjianActivity" android:screenOrientation="portrait"
            android:exported="false" />
        <activity
            android:name=".DashboardActivity" android:screenOrientation="portrait"
            android:exported="false" />
        <activity
            android:name=".LoginActivity" android:screenOrientation="portrait"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>

```

- **Com.example.cbt/model/soal.kt**

```

package com.example.cbt.model

data class Soal(
    val pertanyaan: String,
    val pilihanA: String, val pilihanB: String, val pilihanC:
    String, val pilihanD: String, val jawabanBenar: String
)

```

- **Com.example.cbt/LoginActivity.kt**

```

package com.example.cbt

import android.content.Intent
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

```

```

class LoginActivity : AppCompatActivity() {

    private lateinit var etUsername: EditText
    private lateinit var etPassword: EditText
    private lateinit var btnLogin: Button

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_login)

        etUsername = findViewById(R.id.etUsername)
        etPassword = findViewById(R.id.etPassword)
        btnLogin = findViewById(R.id.btnLogin)

        btnLogin.setOnClickListener {

            val username = etUsername.text.toString().trim()
            val password = etPassword.text.toString().trim()

            if (username.isEmpty() || password.isEmpty()) {

                Toast.makeText(
                    this,
                    "Username dan password wajib diisi",
                    Toast.LENGTH_SHORT
                ).show()

            } else {

                val intent = Intent(this, DashboardActivity::class.java)
                intent.putExtra("username", username)
                startActivity(intent)
                finish()
            }
        }
    }
}

```

- **Res/layout/activity_login.xml**

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android

```

```

android:layout_width="match_parent" android:layout_height="match_parent"
android:orientation="vertical" android:gravity="center" android:padding="24dp"
android:background="#0B132B">
    <TextView
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:text="CBT
        ONLINE" android:textColor="#FFFFFF" android:textSize="30sp"
        android:textStyle="bold" android:gravity="center" />
    <TextView
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:text="Computer Based Test" android:textColor="#B0BEC5"
        android:textSize="16sp" android:gravity="center"
        android:layout_marginBottom="32dp" />
    <EditText android:id="@+id/etUsername" android:layout_width="match_parent"
        android:layout_height="wrap_content"

        android:hint="Masukkan NIM / NIS"
        android:background="#FFFFFF"
        android:padding="14dp"
        android:layout_marginBottom="12dp" />
    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Masukkan Password / Token"
        android:inputType="textPassword"
        android:background="#FFFFFF"
        android:padding="14dp"
        android:layout_marginBottom="20dp" />
    <Button
        android:id="@+id/btnLogin"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Masuk CBT"
        android:textAllCaps="false"
        android:textSize="16sp"
        android:backgroundTint="#1C77C3"
        android:textColor="#FFFFFF" />
</LinearLayout>

```

- **Com.example.cbt/DashboardActivity.kt**

```
package com.example.cbt
```

```

import android.content.Intent
import android.os.Bundle
import android.widget.Button
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity
class DashboardActivity : AppCompatActivity() {
    private lateinit var tvPeserta: TextView
    private lateinit var btnMulai: Button
    override fun onCreate(savedInstanceState: Bundle?)
    { super.onCreate(savedInstanceState)
      setContentView(R.layout.activity_dashboard)
      tvPeserta = findViewById(R.id.tvPeserta)
      btnMulai = findViewById(R.id.btnMulai)

      val username = intent.getStringExtra("username") ?: "Peserta"
      tvPeserta.text = "Selamat Datang, $username"
      btnMulai.setOnClickListener {
          startActivity(Intent(this, UjianActivity::class.java))
      }
    }
}

```

- **Res/layout/activity_dashboard.xml**

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    android:background="#F4F6F9">
    <TextView
        android:id="@+id/tvPeserta"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Peserta"
        android:textSize="22sp"
        android:textStyle="bold"
        android:textColor="#0B132B"
        android:layout_marginBottom="20dp" />
    <TextView
        android:layout_width="match_parent"

```

```

        android:layout_height="wrap_content"
        android:text="Informasi Ujian"
        android:textSize="20sp"
        android:textStyle="bold"

        android:textColor="#1C77C3"
        android:layout_marginBottom="10dp" />
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Mata Kuliah: Sistem Operasi&#10;Jumlah Soal: 5&#10;Waktu: 10
Menit&#10;Jenis: Pilihan Ganda"
    android:textSize="16sp"
    android:textColor="#333333"
    android:background="#FFFFFF"
    android:padding="16dp"
    android:layout_marginBottom="30dp" />
<Button
    android:id="@+id/btnMulai"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Mulai Ujian"
    android:textAllCaps="false"
    android:textSize="17sp"
    android:backgroundTint="#0B132B"
    android:textColor="#FFFFFF" />
</LinearLayout>

```

- **Com.example.cbt/UjianActivity.kt**

```

package com.example.cbt

import android.app.AlertDialog
import android.os.Bundle
import android.os.CountDownTimer
import android.view.View
import android.view.WindowManager
import android.widget.*
import androidx.activity.OnBackPressedCallback
import androidx.appcompat.app.AppCompatActivity
import com.example.cbt.model.Solal

class UjianActivity : AppCompatActivity() {

```

```
private lateinit var tvTimer: TextView
private lateinit var tvNomorSoal: TextView
private lateinit var tvPertanyaan: TextView
```

```
private lateinit var rgPilihan: RadioGroup
```

```
private lateinit var rbA: RadioButton
private lateinit var rbB: RadioButton
private lateinit var rbC: RadioButton
private lateinit var rbD: RadioButton
```

```
private lateinit var btnSebelumnya: Button
private lateinit var btnSelanjutnya: Button
private lateinit var btnSelesai: Button
```

```
private var nomorSoal = 0
private var pelanggaran = 0
```

```
private val jawabanPeserta = HashMap<Int, String>()
```

```
private val daftarSoal = arrayListOf(
```

```
    Soal(
        "Apa fungsi utama sistem operasi?",
        "Mengatur perangkat keras dan perangkat lunak",
        "Membuat desain grafis",
        "Menghapus seluruh data",
        "Mempercepat internet",
        "A"
    ),
```

```
    Soal(
        "RAM termasuk jenis memori apa?",
        "Memori sekunder",
        "Memori utama",
        "Memori eksternal",
        "Memori cadangan",
        "B"
    ),
```

```
    Soal(
        "CPU sering disebut sebagai?",
        "Otak komputer",
```

```

        "Penyimpanan data",
        "Layar komputer",
        "Perangkat input",
        "A "
    ),

    Soal(
        "Contoh perangkat input adalah?",
        "Monitor",
        "Printer",
        "Keyboard",
        "Speaker",
        "C"
    ),

    Soal(
        "Ekstensi file Android Package adalah?",
        ".exe",
        ".docx",
        ".apk",
        ".pdf",
        "C"
    )
)

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)

    // Mencegah screenshot
    window.setFlags(
        WindowManager.LayoutParams.FLAG_SECURE,
        WindowManager.LayoutParams.FLAG_SECURE
    )

    setContentView(R.layout.activity_ujian)

    initView()
    aktifkanModeAman()
    mulaiTimer()
    tampilkanSoal()

    btnSebelumnya.setOnClickListener {

        simpanJawaban()
    }
}

```

```

        if (nomorSoal > 0) {
            nomorSoal--
            tampilkanSoal()
        }
    }

    btnSelanjutnya.setOnClickListener {

        simpanJawaban()

        if (nomorSoal < daftarSoal.size - 1) {
            nomorSoal++
            tampilkanSoal()
        }
    }

    btnSelesai.setOnClickListener {
        simpanJawaban()
        konfirmasiSelesai()
    }
}

private fun initView() {

    tvTimer = findViewById(R.id.tvTimer)
    tvNomorSoal = findViewById(R.id.tvNomorSoal)
    tvPertanyaan = findViewById(R.id.tvPertanyaan)

    rgPilihan = findViewById(R.id.rgPilihan)

    rbA = findViewById(R.id.rbA)
    rbB = findViewById(R.id.rbB)
    rbC = findViewById(R.id.rbC)
    rbD = findViewById(R.id.rbD)

    btnSebelumnya = findViewById(R.id.btnSebelumnya)
    btnSelanjutnya = findViewById(R.id.btnSelanjutnya)
    btnSelesai = findViewById(R.id.btnSelesai)
}

private fun aktifkanModeAman() {

    onBackPressedDispatcher.addCallback(

```

```

        this,
        object : OnBackPressedCallback(true) {

            override fun handleOnBackPressed() {

                Toast.makeText(
                    this@UjianActivity,
                    "Tombol kembali dinonaktifkan selama ujian",
                    Toast.LENGTH_SHORT
                ).show()
            }
        }
    )

    window.decorView.systemUiVisibility =
        View.SYSTEM_UI_FLAG_FULLSCREEN or
        View.SYSTEM_UI_FLAG_HIDE_NAVIGATION or
        View.SYSTEM_UI_FLAG_IMMERSIVE_STICKY

    try {
        startLockTask()
    } catch (e: Exception) {
        e.printStackTrace()
    }
}

private fun tampilkanSoal() {

    val soal = daftarSoal[nomorSoal]

    tvNomorSoal.text =
        "Soal ${nomorSoal + 1} dari ${daftarSoal.size}"

    tvPertanyaan.text = soal.pertanyaan

    rbA.text = "A. ${soal.pilihanA}"
    rbB.text = "B. ${soal.pilihanB}"
    rbC.text = "C. ${soal.pilihanC}"
    rbD.text = "D. ${soal.pilihanD}"

    rgPilihan.clearCheck()

    when (jawabanPeserta[nomorSoal]) {

```

```

        "A" -> rbA.isChecked = true
        "B" -> rbB.isChecked = true
        "C" -> rbC.isChecked = true
        "D" -> rbD.isChecked = true
    }
}

private fun simpanJawaban() {

    when (rgPilihan.checkedRadioButtonId) {

        R.id.rbA -> jawabanPeserta[nomorSoal] = "A"
        R.id.rbB -> jawabanPeserta[nomorSoal] = "B"
        R.id.rbC -> jawabanPeserta[nomorSoal] = "C"
        R.id.rbD -> jawabanPeserta[nomorSoal] = "D"
    }
}

private fun mulaiTimer() {

    val waktuUjian = 10 * 60 * 1000L

    object : CountDownTimer(waktuUjian, 1000) {

        override fun onTick(millisUntilFinished: Long) {

            val menit = millisUntilFinished / 1000 / 60
            val detik = millisUntilFinished / 1000 % 60

            tvTimer.text =
                String.format("Sisa Waktu: %02d:%02d", menit, detik)
        }

        override fun onFinish() {
            hitungNilai()
        }

    }.start()
}

private fun konfirmasiSelesai() {

    AlertDialog.Builder(this)
        .setTitle("Selesai Ujian")

```

```

        .setMessage("Apakah Anda yakin ingin menyelesaikan ujian?")
        .setPositiveButton("Ya") { _, _ ->
            hitungNilai()
        }
        .setNegativeButton("Tidak", null)
        .show()
    }

```

```

private fun hitungNilai() {

    var benar = 0

    for (i in daftarSoal.indices) {

        if (jawabanPeserta[i] == daftarSoal[i].jawabanBenar) {
            benar++
        }
    }

```

```

    val nilai = benar * 100 / daftarSoal.size

```

```

    try {
        stopLockTask()
    } catch (e: Exception) {
        e.printStackTrace()
    }

```

```

AlertDialog.Builder(this)
    .setTitle("Hasil Ujian")
    .setMessage(
        "Jawaban Benar: $benar dari ${daftarSoal.size}\n" +
        "Nilai: $nilai\n" +
        "Pelanggaran keluar aplikasi: $pelanggaran"
    )
    .setCancelable(false)
    .setPositiveButton("Selesai") { _, _ ->
        finish()
    }
    .show()
}

```

```

override fun onPause() {
    super.onPause()
}

```

```

        pelanggaran++

        if (pelanggaran >= 2) {

            Toast.makeText(
                this,
                "Terdeteksi keluar dari aplikasi CBT",
                Toast.LENGTH_LONG
            ).show()
        }
    }

    override fun onWindowFocusChanged(hasFocus: Boolean) {
        super.onWindowFocusChanged(hasFocus)

        if (!hasFocus) {

            pelanggaran++

        } else {

            window.decorView.systemUiVisibility =
                View.SYSTEM_UI_FLAG_FULLSCREEN or
                View.SYSTEM_UI_FLAG_HIDE_NAVIGATION or
                View.SYSTEM_UI_FLAG_IMMERSIVE_STICKY
        }
    }
}

```

- **Res/layout/activity_ujian.xml**

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/layoutUjian"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="18dp"
    android:background="#F4F6F9">

    <TextView
        android:id="@+id/tvTimer"
        android:layout_width="match_parent"

```

```
android:layout_height="wrap_content"
android:text="Sisa Waktu: 10:00"
android:textSize="18sp"
android:textStyle="bold"
android:textColor="#D32F2F"
android:gravity="end"
android:layout_marginBottom="16dp" />
```

```
<TextView
    android:id="@+id/tvNomorSoal"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Soal 1 dari 5"
    android:textSize="18sp"
    android:textStyle="bold"
    android:textColor="#0B132B"
    android:layout_marginBottom="12dp" />
```

```
<TextView
    android:id="@+id/tvPertanyaan"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Pertanyaan soal"
    android:textSize="18sp"
    android:textColor="#222222"
    android:background="#FFFFFF"
    android:padding="16dp"
    android:layout_marginBottom="18dp" />
```

```
<RadioGroup
    android:id="@+id/rgPilihan"
    android:layout_width="match_parent"
    android:layout_height="wrap_content">
```

```
<RadioButton
    android:id="@+id/rbA"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="A"
    android:textSize="16sp"
    android:padding="10dp" />
```

```
<RadioButton
    android:id="@+id/rbB"
```

```
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="B"
    android:textSize="16sp"
    android:padding="10dp" />
```

```
<RadioButton
    android:id="@+id/rbC"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="C"
    android:textSize="16sp"
    android:padding="10dp" />
```

```
<RadioButton
    android:id="@+id/rbD"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="D"
    android:textSize="16sp"
    android:padding="10dp" />
```

```
</RadioGroup>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginTop="24dp">
```

```
<Button
    android:id="@+id/btnSebelumnya"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:text="Sebelumnya"
    android:textAllCaps="false" />
```

```
<Button
    android:id="@+id/btnSelanjutnya"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:text="Selanjutnya"
    android:textAllCaps="false"
```

```
        android:layout_marginStart="10dp" />
</LinearLayout>
```

```
<Button
    android:id="@+id/btnSelesai"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Selesai Ujian"
    android:textAllCaps="false"
    android:backgroundTint="#D32F2F"
    android:textColor="#FFFFFFF"
    android:layout_marginTop="14dp" />
```

```
</LinearLayout>
```

3.kemudian pada bagian **build.gradle.kts(module: app)**, ubah programnya seperti ini:

- Program

```
plugins {
    alias(libs.plugins.android.application)
    alias(libs.plugins.kotlin.android)
}

android {
    namespace = "com.example.cbt"
    compileSdk = 36

    defaultConfig {
        applicationId = "com.example.cbt"
        minSdk = 24
        targetSdk = 36
        versionCode = 1
        versionName = "1.0"

        testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"
    }

    buildTypes {
        release {
            isMinifyEnabled = false
            proguardFiles(
                getDefaultProguardFile("proguard-android-optimize.txt"),
```

```

        "proguard-rules.pro"
    )
}
}
compileOptions {
    sourceCompatibility = JavaVersion.VERSION_11
    targetCompatibility = JavaVersion.VERSION_11
}
kotlinOptions {
    jvmTarget = "11"
}
}

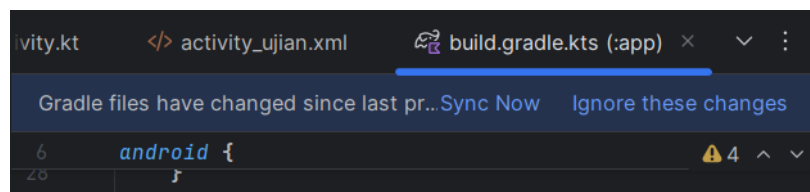
dependencies {

    implementation("androidx.core:core-ktx:1.15.0")
    implementation("androidx.appcompat:appcompat:1.7.0")
    implementation("androidx.activity:activity-ktx:1.9.3")

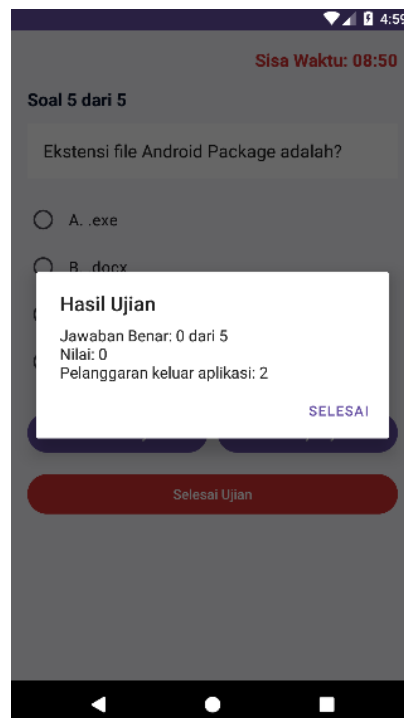
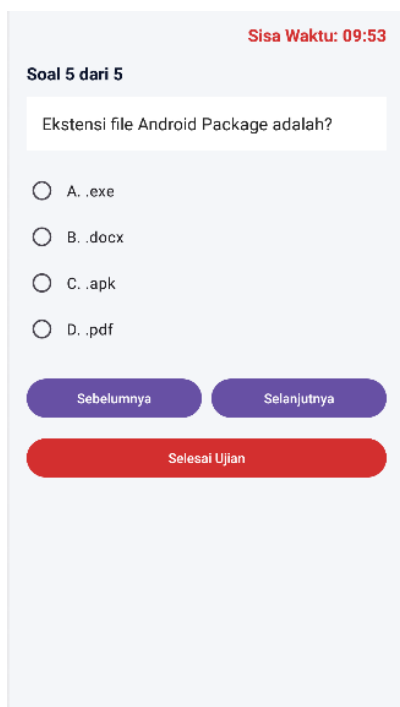
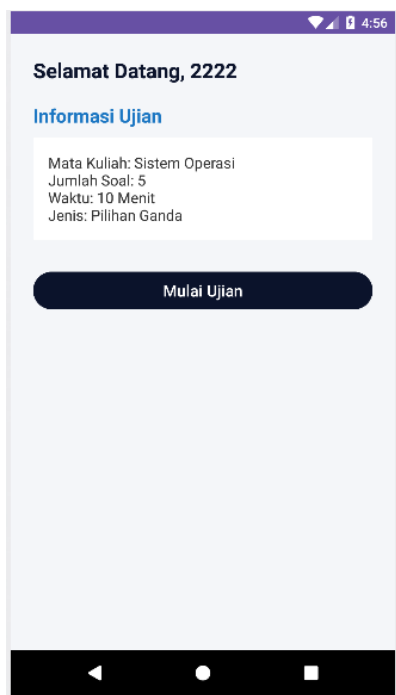
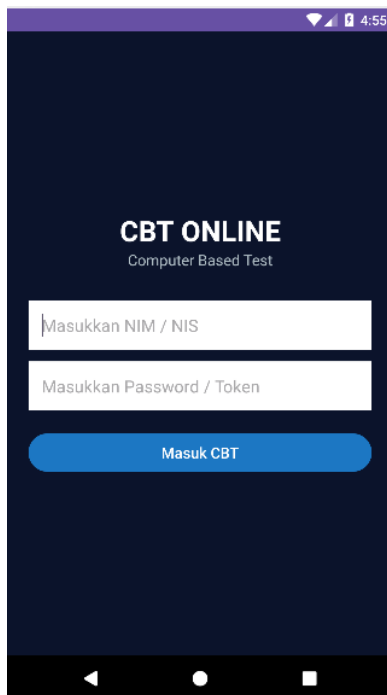
    implementation(libs.androidx.core.ktx)
    implementation(libs.androidx.appcompat)
    implementation(libs.material)
    implementation(libs.androidx.activity)
    implementation(libs.androidx.constraintlayout)
    testImplementation(libs.junit)
    androidTestImplementation(libs.androidx.junit)
    androidTestImplementation(libs.androidx.espresso.core)
}

```

- Kemudian sync now



4.hasil/output dari program:

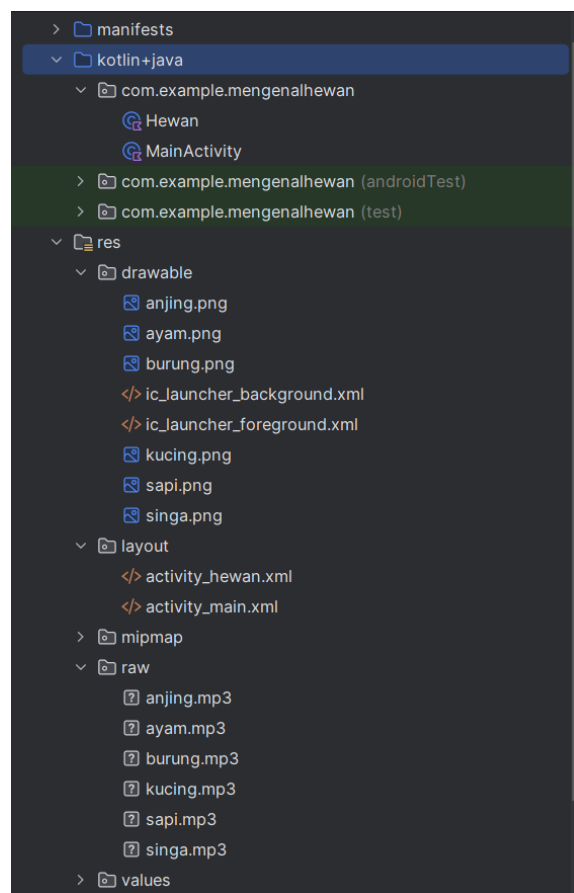


RANCANGAN APLIKASI CBT PROFESIONAL

5. Buat Project baru dengan create new project kemudian pilih empty views activity kemudian isi

- Name: mengenalhewan
- Packagename: com.example.mengenalhewan
- Language: Kotlin
- Klik selesai

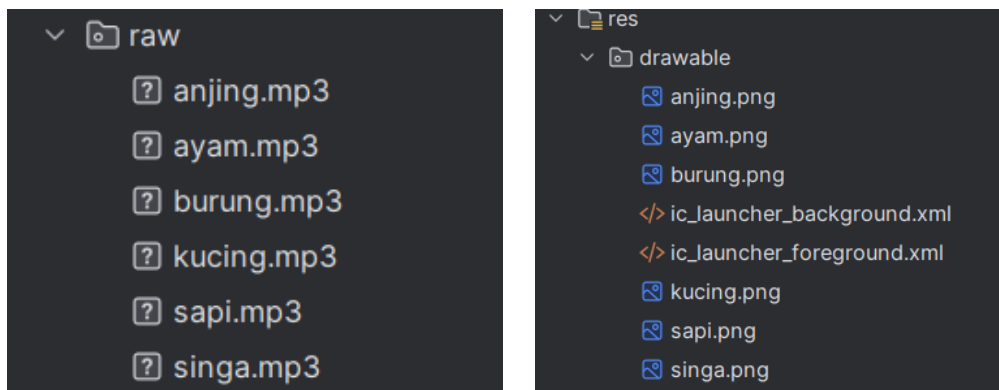
Kemudian buat struktur foldernya seperti ini:



Cara membuat folder raw:

Klik kanan res > New > Android Resource Directory > Resource type: raw

6. kemudian tambahkan mp3 hewan pada folder raw dan gambar hewannya pada drawable



7. kemudian tambahkan kode program berikut:

- Hewan.kt

```
package com.example.mengenalhewan
```

```
data class
```

```
Hewan( val nama:
```

```
    String, val
```

```
    gambar: Int, val
```

```
    suara: Int
```

```
)
```

- MainActivity.kt

```
package com.example.mengenalhewan
```

```
import android.media.MediaPlayer
```

```
import android.os.Bundle
```

```
import android.view.Gravity
```

```
import android.widget.GridLayout
```

```
import android.widget.ImageView
```

```
import android.widget.LinearLayout
```

```
import android.widget.TextView
```

```
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {
```

```
    private var mediaPlayer: MediaPlayer? = null
```

```
    private val daftarHewan = listOf(
```

```

        Hewan("Kucing", R.drawable.kucing, R.raw.kucing),
        Hewan("Anjing", R.drawable.anjing, R.raw.anjing),
        Hewan("Sapi", R.drawable.sapi, R.raw.sapi),
        Hewan("Ayam", R.drawable.ayam, R.raw.ayam),
        Hewan("Burung", R.drawable.burung, R.raw.burung),
        Hewan("Singa", R.drawable.singa, R.raw.singa)
    )

```

```

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)

    setContentView(R.layout.activity_main)

    val gridHewan = findViewById<GridLayout>(R.id.gridHewan)

    daftarHewan.forEach { hewan ->
        val card = buatCardHewan(hewan)
        gridHewan.addView(card)
    }
}

```

```

private fun buatCardHewan(hewan: Hewan): LinearLayout {

    val card = LinearLayout(this)

    card.orientation = LinearLayout.VERTICAL
    card.gravity = Gravity.CENTER
    card.setPadding(16, 16, 16, 16)

    card.setBackgroundResource(
        android.R.drawable.dialog_holo_light_frame
    )

    val params = GridLayout.LayoutParams()

    params.width = 0
    params.height = LinearLayout.LayoutParams.WRAP_CONTENT
    params.columnSpec = GridLayout.spec(GridLayout.UNDEFINED, 1f)

    params.setMargins(12, 12, 12, 12)

    card.layoutParams = params

    val imageView = ImageView(this)

```

```

imageView.setImageResource(hewan.gambar)

imageView.layoutParams =
    LinearLayout.LayoutParams(220, 220)

imageView.scaleType = ImageView.ScaleType.CENTER_CROP

val textView = TextView(this)

textView.text = hewan.nama
textView.textSize = 20f

textView.setTextColor(
    android.graphics.Color.parseColor("#1A237E")
)

textView.gravity = Gravity.CENTER
textView.setPadding(0, 12, 0, 8)

card.addView(imageView)
card.addView(textView)

card.setOnClickListener {
    putarSuara(hewan.suara)
}

return card
}

private fun putarSuara(suara: Int) {

    mediaPlayer?.stop()
    mediaPlayer?.release()

    mediaPlayer = MediaPlayer.create(this, suara)

    mediaPlayer?.start()

    mediaPlayer?.setOnCompletionListener {

        it.release()
        mediaPlayer = null
    }
}

```

```

    }

    override fun onDestroy() {
        super.onDestroy()

        mediaPlayer?.release()
        mediaPlayer = null
    }
}

```

- Activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#F4F8FB">
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">
        <TextView
            android:id="@+id/tvJudul"

            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Mengenal Nama Hewan"
            android:textSize="26sp"
            android:textStyle="bold"
            android:textColor="#1A237E"
            android:gravity="center"
            android:padding="12dp" />
        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Klik gambar hewan untuk mendengar suaranya"
            android:textSize="16sp"
            android:textColor="#333333"
            android:gravity="center"
            android:paddingBottom="16dp" />
        <GridLayout
            android:id="@+id/gridHewan"

```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:columnCount="2"
        android:useDefaultMargins="true">
    </GridLayout>
</LinearLayout>
</ScrollView>
```

8.output programnya seperti ini

